

1. Administrative & Study Identification

Brief Study Title: Study to Evaluate Efficacy and Safety of Inclisiran in Adolescents With Homozygous Familial Hypercholesterolemia **Full Study Title:** Two Part (Double-blind Inclisiran Versus Placebo [Year 1] Followed by Open-label Inclisiran [Year 2]) Randomized Multicenter Study to Evaluate Safety, Tolerability, and Efficacy of Inclisiran in Adolescents With Homozygous Familial Hypercholesterolemia and Elevated LDL-cholesterol **Short Title/Acronym:** ORION-13 **Posting ID:** 56129907882111838 **Protocol Number:** CKJX839C12302 (EudraCT: 2020-002755-38) **NCT Number:** NCT04659863 **Trial Status:** COMPLETED **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** Inclisiran (Leqvio / KJX839) | **ATC Code:** C10AX16 **Phase of Development:** Phase III **Medical Monitor:** Novartis Clinical Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy (percentage change in LDL-C) of inclisiran compared to placebo from baseline to Day 330 in adolescents with homozygous familial hypercholesterolemia (HoFH) and elevated LDL-C. **Secondary Objectives:** To evaluate the absolute and percentage changes in other lipid parameters (Apo A1, PCSK9, non-HDL-C, total cholesterol, triglycerides, HDL-C, Lp(a)) from baseline up to Day 720, and to assess long-term safety and tolerability.

2.2 Background & Rationale HoFH is a rare, life-threatening inherited genetic disorder characterized by severely elevated LDL-C levels from birth, leading to early-onset atherosclerotic cardiovascular disease (ASCVD). Despite maximal standard-of-care lipid-lowering therapies (e.g., statins, ezetimibe), many pediatric patients fail to reach LDL-C goals. Inclisiran is a first-in-class small interfering RNA (siRNA) that inhibits hepatic PCSK9 synthesis, thereby increasing LDL-C uptake. While proven effective in adults, this study addresses the critical unmet need to evaluate inclisiran's efficacy and safety in adolescent patients to reduce lifelong cardiovascular risk.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled (during Year 1) **Blinding:** Double-blind (Year 1), followed by Open-label (Year 2) **Randomization:** 2:1 (Inclisiran to Placebo in Year 1)

3.2 Planned Enrollment & Duration Trial Status: COMPLETED **Target**
\$N\$: 13 adolescent participants **Number of Sites:** 11 global
locations **Estimated Study Start:** 16-Feb-2021 **Estimated Primary**
Completion: 30-Oct-2023

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or female participants ≥ 12 to < 18 years of age at screening. 2. Homozygous Familial Hypercholesterolemia (HoFH) diagnosed by genetic confirmation. 3. Fasting LDL-C > 130 mg/dL (3.4 mmol/L) at screening. 4. On a maximally tolerated dose of statin (investigator's discretion) with or without other lipid-lowering therapy; stable for ≥ 30 days before screening.

4.2 Exclusion Criteria 1. Documented evidence of a null (negative) mutation in both LDLR alleles. 2. Heterozygous familial hypercholesterolemia (HeFH) or secondary hypercholesterolemia (e.g., hypothyroidism, nephrotic syndrome). 3. Active liver disease. 4. Previous treatment with monoclonal antibodies directed towards PCSK9 within 90 days of screening. 5. Treatment with mipomersen or lomitapide within 5 months of screening. 6. Recent and/or planned use of other investigational medicinal products or devices.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Inclisiran sodium 300 mg administered on Days 1, 90, and 270 (Part 1/Year 1). During Part 2 (Year 2), all subjects received open-label inclisiran 300 mg on Days 450 and 630 (placebo subjects transitioned to active drug). **Route of Administration:** Subcutaneous (s.c.) injection. **Duration of Treatment:** 2 years (evaluations up to Day 720).

5.2 Concomitant & Prohibited Therapy Permitted: Stable standard of care background lipid-lowering therapy, including maximally tolerated statins and routine LDL-apheresis (must remain on a stable schedule). **Prohibited:** PCSK9 monoclonal antibodies, mipomersen, lomitapide, and other investigational medicinal products.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Up to Day 0 Informed Consent/ Assent, Medical History, Fasting Lipid Profile, Genetic Screening
Baseline	Day 1	Randomization, First s.c. Injection	
(Inclisiran or Placebo), Baseline Labs	Interim	Days 90, 270	
s.c. Injections, Efficacy (Lipid) Assessment, Safety Labs			

Primary Efficacy | Day 330 | Primary Endpoint Evaluation (LDL-C percentage change) | | **End of Study** | Days 450, 630, 720 | Open-label Dosing (Days 450, 630), Final Efficacy & Safety Evaluation (Day 720) |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Percentage change in low-density lipoprotein cholesterol (LDL-C) from baseline to Day 330 (Part 1/Year 1). **Measurement Method:** Central Laboratory Analysis of fasting blood samples.

7.2 Secondary & Exploratory Endpoints - Time-adjusted Percent Change in LDL-C From Baseline After Day 90 and up to Day 330 (Part 1/Year 1). - Percent Change in LDL-C From Baseline up to Day 720. - Absolute Change in LDL-C From Baseline up to Day 720. - Absolute and percentage change in PCSK9, Apolipoprotein A1 (Apo A1), non-HDL-C, total cholesterol, triglycerides, HDL-C, and Lipoprotein(a) [Lp(a)] from baseline up to Day 720. - Incidence of treatment-emergent adverse events (TEAEs) and serious adverse events (SAEs).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Collection of AEs, injection-site reactions, and clinical laboratory abnormalities (e.g., hepatic markers) at all visits. **Serious Adverse Events (SAEs):** Standard expedited reporting to the Sponsor and Health Authorities within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee appointed to periodically review unblinded safety and tolerability data.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy modeling; Safety Set for all AE/SAE evaluations. **Sample Size Justification:** As HoFH is an ultra-rare pediatric condition, a sample size of ~13 participants is deemed sufficient to detect meaningful reductions and is standard for this indication. **Handling of Missing Data:** Handled via mixed-effect model for repeated measures (MMRM) and/or Multiple Imputation for continuous lipid parameters at Day 330.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Scheduled onsite and remote monitoring visits to

review protocol adherence, investigational product accountability, and source document verification. **Audit Trail:** Validated electronic Data Capture (eDC) via eCRF locking, with electronic signatures and active queries for data clarification.

11. Study Identifiers & Governance

Primary ID: CKJX839C12302 **Registry Number:** NCT04659863 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** ORION-13 **Official Regulatory Title:** Two Part (Double-blind Inclisiran Versus Placebo [Year 1] Followed by Open-label Inclisiran [Year 2]) Randomized Multicenter Study to Evaluate Safety, Tolerability, and Efficacy of Inclisiran in Adolescents With Homozygous Familial Hypercholesterolemia and Elevated LDL-cholesterol **EMA URL:** https://www.ema.europa.eu/en/documents/product-information/leqvio-epar-product-information_en.pdf

12. Clinical Framework (Familial Hypercholesterolemia Specific)

Primary Condition: Homozygous Familial Hypercholesterolemia (HoFH) **MeSH Code:** D000090542 **SNOMED ID:** 238078005 **Diagnosis Threshold:** Genetic Confirmation & Fasting LDL-C ≥ 130 mg/dL (3.4 mmol/L) **Medical Methodology:** mRNA targeted therapy (PCSK9-targeted small interfering RNA) **Optimization Target:** Significant and sustained reduction in LDL-C and atherogenic lipoproteins

13. Study Procedures & Methodology

Management Pathway: Subcutaneous siRNA injections alongside optimized standard-of-care lipid-lowering therapies (including stable apheresis). **Education Protocol (HCP):** Investigator site training on specific subcutaneous injection protocols and pediatric/adolescent clinical evaluation. **Education Protocol (Patient):** Age-appropriate assent materials and disease education counseling alongside caregivers. **Quality Audit Mechanism:** Central laboratory lipid analyses and standardized clinical monitoring by Clinical Research Associates.

14. Observation & Follow-Up Schedule

Total Duration: 720 Days (approx. 2 years) **Onsite Visit Cadence:** Day 1, 90, 270, 330, 450, 630, and 720 **Remote Monitoring Frequency:** Standard interval remote monitoring for safety tracking **Checklist Review Interval:** AEs, concomitant medications, and procedural assessments reviewed at every visit

15. Sign-off & Approval

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					
Statistician	[Name]	_____	[DD-MMM-YYYY]					